MARRIAGE & DIVORCE RISK ANALYSIS

SQL + Python Exploratory Data Analysis Project

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Executive Summary

This report presents a comprehensive exploratory data analysis of a Marriage & Divorce dataset consisting of 100 observations and 30 relationship-related features. The primary objective is to identify which emotional, social, economic, and family-related variables are most strongly associated with divorce probability.

Using SQL for structured data exploration and Python (Pandas, NumPy, Matplotlib) for visualization and feature engineering, the analysis categorized couples into Low, Medium, and High Risk groups and developed a custom Relationship Health Score to evaluate overall relationship quality.

Key findings reveal that approximately 40% of couples fall into the high-risk category, and that education level, financial stability, and addictive behaviors demonstrate the most measurable associations with divorce probability.

Dataset Overview

The dataset was sourced from peer-reviewed academic research on divorce prediction and family counseling systems. It contains demographic, behavioral, social, and relationship-related variables.

Dataset Summary Statistics

Total Records	100
Total Features	30
Avg. Divorce Probability	2.067
Min. Divorce Probability	1.010
Max. Divorce Probability	2.990

Key Features Analyzed

- Love, Commitment, Loyalty
- Mental Health & Self-Confidence
- Good Income & Economic Similarity
- Religion Compatibility & Family Approval
- Addiction & Previous Marriage
- Family Divorce History & Age Gap
- Education Level

Methodology & Project Workflow

The project followed a structured 8-stage analytics pipeline:

Stage	Activity	Tools Used
1	Data Collection	CSV / Raw Dataset
2	Data Cleaning	Python / Pandas
3	SQL Analysis	SQL (MySQL/SQLite)
4	Exploratory Data Analysis	Python / Pandas / NumPy
5	Feature Engineering	Python / Custom Scoring
6	Data Visualization	Matplotlib
7	Insight Generation	Statistical Analysis
8	Reporting	Markdown / Word

Risk Categorization

To improve interpretability and clinical usefulness, divorce probability scores were segmented into three distinct risk categories:

Risk Category	Probability Range	Record Count	% of Dataset
Low Risk	< 1.7	29	29%
Medium Risk	1.7 – 2.3	31	31%
High Risk	> 2.3	40	40%

High-risk couples represent the single largest group at 40% of observations, indicating that a significant portion of the dataset exhibits elevated divorce probability. This distribution underscores the practical value of targeted counseling interventions for at-risk couples.

Feature Engineering

Relationship Health Score

A custom composite metric was engineered to provide a single interpretable measure of overall relationship quality:

```
Health Score = (Love + Commitment + Loyalty + Mental Health + Self-Confidence) / 5
```

Average Relationship Health Score across the dataset: 67.15 / 100

Importantly, the analysis found that higher health scores do not consistently correspond to lower divorce probability, suggesting that relationship quality alone is insufficient to predict marital outcomes — external and structural factors play a significant role.

Emotional Factor Analysis

Average scores for the five core emotional variables measured across all 100 respondents:

Emotional Factor	Average Score	Rank
Mental Health	74.4	1st
Self-Confidence	71.4	2nd
Commitment	68.7	3rd
Loyalty	62.0	4th
Love	59.3	5th

Mental health emerged as the highest-scoring emotional factor, followed closely by self-confidence. Love, while emotionally central to relationships, scored lowest on average — indicating substantial variability in emotional attachment across the sample.

Correlation Analysis

Top features by correlation coefficient with Divorce Probability:

Feature	Correlation	Direction	Interpretation
Education	-0.32	Negative	Higher education reduces divorce risk
Good Income	-0.18	Negative	Financial stability improves outcomes
Addiction	+0.12	Positive	Addictive behavior increases risk
Love	+0.10	Positive	Complex non-linear relationship
Age Gap	+0.10	Positive	Larger gaps slightly elevate risk

Education demonstrates the strongest correlation with divorce probability and is the most actionable structural factor identified. Financial stability provides meaningful protective effects, while addiction serves as the most influential behavioral risk factor.

Key Business Insights & Findings

Finding 1 — High-Risk Population Prevalence

40% of couples fall into the high-risk category. This group should be prioritized for marriage counseling and early-intervention programs.

Finding 2 — Education as Protective Factor

Education demonstrates the strongest negative association with divorce probability (-0.32). Programs promoting educational attainment may contribute to long-term relationship stability.

Finding 3 — Financial Stability Matters

Good income shows a moderate negative relationship with divorce risk (-0.18), reinforcing the importance of economic counseling and financial planning for couples.

Finding 4 — Addiction as a Risk Amplifier

Addiction carries a measurable positive correlation (+0.12) with divorce probability, consistent with existing clinical research on addictive behaviors and marital instability.

Finding 5 — Multifactorial Nature of Divorce Risk

Relationship success cannot be explained by any single emotional characteristic. Love and commitment are important but insufficient on their own — social, behavioral, and economic variables interact simultaneously to determine outcomes.

SQL Analysis Summary

The SQL component of the project was organized across three complexity tiers:

Tier	Techniques Used
Beginner	COUNT, AVG, MIN, MAX, ORDER BY, LIMIT
Intermediate	CASE statements, risk categorization, GROUP BY, aggregate functions
Advanced	Subqueries, window functions, ranking, custom health score calculation

Example SQL Query — Average Divorce Probability:

```
SELECT AVG(`Divorce Probability`) AS avg_divorce_probability FROM
marriage_divorce_db;
```

Visualizations Produced

The following visual analyses were generated during the Python EDA phase:

- Divorce Probability Distribution (Histogram)
- Divorce Risk Category Distribution (Bar Chart)
- Average Divorce Probability by Risk Group (Grouped Bar)
- Relationship Health Score vs. Divorce Probability (Scatter Plot)
- Love vs. Divorce Probability (Scatter Plot)
- Commitment vs. Divorce Probability (Scatter Plot)
- Relationship Factor Comparison (Radar / Bar Chart)
- Correlation Analysis Heatmap

Future Improvements & Roadmap

Priority	Enhancement	Expected Value
High	Predictive Modeling (Random Forest, XGBoost)	Quantify feature importance & forecast risk
High	SHAP Explainability	Interpret individual prediction drivers
Medium	Interactive Dashboards (Plotly / Streamlit)	Enable real-time exploration for counselors
Medium	Larger-scale datasets	Improve statistical power and generalizability

Priority	Enhancement	Expected Value
Low	Longitudinal tracking	Monitor relationship changes over time

Conclusion

This project demonstrates a complete, production-quality data analytics workflow applied to a real-world social science dataset. By combining SQL's structured query power with Python's visualization and feature engineering capabilities, the analysis successfully identified actionable risk factors associated with divorce probability.

The findings confirm that divorce outcomes are multifactorial and cannot be attributed to any single cause. Education, income, addiction, and the interplay of social and behavioral dynamics all contribute meaningfully to marital stability. The Relationship Health Score provides a practical summary metric that could be incorporated into counseling assessment tools.

Future extensions into machine learning — particularly gradient boosting models paired with SHAP interpretability — would substantially enhance the predictive and explanatory power of this analysis, creating genuine value for practitioners in family counseling and social policy.

References

Mousavi, S. M. H., MiriNezhad, S. Y., & Lyashenko, V. (2017). An Evolutionary-Based Adaptive Neuro-Fuzzy Expert System as a Family Counselor Before Marriage with the Aim of Divorce Rate Reduction. Dataset source and theoretical foundation for this analysis.

Tools & Libraries: Python 3.x, Pandas >= 2.0.0, NumPy >= 1.24.0, Matplotlib >= 3.7.0, Jupyter Notebook, VS Code, Git/GitHub, SQL (MySQL / SQLite).